Yanan He

New Haven, CT   https://echoo113.github.io

*I study why AI **works** when it works, why it **fails** when it doesn't, and how understanding those differences can help us **build better models**.*

EDUCATION

Yale University, New Haven, CT — Aug 2026 – May 2028
M.S. in Computer Science (*Fully Funded*)

Purdue University, West Lafayette, IN — Aug 2023 – May 2026
B.S. in Computer Science & Mathematics-Statistics — GPA: 3.96/4.0 | *Graduated with Distinction*

RESEARCH EXPERIENCE

NLP Lab, Yale University — *New Haven, CT*
Research Assistant | Advisor: Arman Cohan — Apr 2026 – Present

- **Model Behavior, Interpretability & Post-Training** – investigating the mechanisms underlying **subliminal learning** and using these insights to improve how language models learn during post-training.
- **LLM-as-a-Judge for Deep Research Agents** – developed REFLECT, a fine-grained benchmark for evaluating judge reliability across agent reasoning, tool use, and final reports [1]

Graph Lab, Stony Brook University — *Stony Brook, NY*
Research Assistant | Advisor: Tengfei Ma — May 2025 – Apr 2026

- **Time-Series Modeling** – mitigated **latent prediction collapse** in JEPA with structured codebooks [2]; detected **causal residual shifts** via multi-scale structural consistency [3].
- **LLM Reasoning in Geometric Space** – guided multi-step reasoning with **hyperbolic latent signals** [4].

SELECTED PUBLICATIONS

1. Time to REFLECT: Can We Trust LLM Judges for Evidence-based Research Agents? — * Equal contribution.
   Leyao Wang*, **Yanan He***, Peng Chen, Asaf Yehudai, Yixin Liu, Rex Ying, Michal Shmueli-Scheuer, Arman Cohan
   arXiv:2605.19196.
2. SC-JEPA: Stabilizing Latent Predictive Learning for Time-Series Anomaly Prediction
   Yanan He, Yunshi Wen, Xin Wang, Tengfei Ma
   SDM 2026; Forecast@ICML26.
3. CAAD: Causality-Aware Multivariate Time Series Anomaly Detection via Multi-Scale Alignment and Structural Causal Consistency
   Xin Wang, Yunshi Wen, **Yanan He**, Haotian Xu, Youlan Zhao, Michel Ferreira Cardia Haddad, Tengfei Ma
   SIGKDD 2026 Research Track.
4. HyperGuide: Hyperbolic Guidance for Efficient Multi-Step Reasoning in Large Language Models
   Yuyu Liu, Haotian Xu, **Yanan He**, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma
   arXiv:2605.24140.

INTERESTS & SKILLS

- **Programming Languages:** Python, Java, C/C++, Assembly, R, SQL, MATLAB, JavaScript, HTML/CSS
- **Languages:** English (Professional), Chinese (Native)